Self-Supervised Feature Representation and Few-Shot Land Cover Classification of Multimodal Remote Sensing Images

Zhixiang Xue, Bing Liu, Anzhu Yu, Xuchu Yu, Pengqiang Zhang, and Xiong Tan

***Abstract*— Although deep learning-based approaches have made significant progress in remote sensing (RS) image classification, the supervised learning paradigm has shortcomings under a limited number of labeled samples, which restricts the classification performance to a great extent. In this article, we investigate an effective self-supervised feature representation (SSFR) architecture for multimodal RS images' few-shot land cover classification. Specifically, we exploit a multiview learning strategy to construct multiple views from multimodal RS images. This method builds several complementary views of the same observed scenes from hyperspectral images or different modalities of RS data. Then, we build the deep feature extractor to learn high-level feature representations from each view via contrastive learning. Contrastive learning aggregates the samples of the same scene while separating samples of different scenes in the latent space, and this process does not require any labeled information. What is more, to learn more robust features from different views, we utilize multitask learning strategy to train the feature extraction network. Finally, a lightweight machine learning method is employed to classify the learned features using a few annotated samples. To further demonstrate the self-supervised feature learning capability of the proposed model, we train the feature representation network in multiple source datasets. Comprehensive feature learning and classification experiments have certified the effectiveness and superiority of the proposed method.**

***Index Terms*— Contrastive learning, feature representation, few-shot classification, multimodal images, multitask learning, multiview learning.**

I. Introduction

ACCURATE land cover classification using remote sensing (RS) data is one of the important components in Earth observation (EO) missions. With the rapid development of RS technology, there are multimodal RS images within the same observed scenes (e.g., hyperspectral images (HSIs) have hundreds of diagnostic spectral bands, very-high-resolution (VHR) images contain precise shape information, and light detection and ranging (LiDAR) data can record the elevation information of observed objects) [1]. Therefore, it is a promising paradigm to utilize rich information contained in multimodal RS data for land cover collaborative classification.

It is undeniable that deep learning (DL) has revolutionized the mode of RS data analysis, especially in the field of HSI classification [2], [3]. In recent years, several representative DL models have been applied in the HSI supervised classification task [4]. Containing multiple convolutional layers, convolutional neural networks (CNNs) can extract spectral, spatial, and spectral–spatial features, respectively, for hyperspectral data classification [5], [6], [7]. As a sequence model, the recurrent neural network (RNN) treats the spectral information of each pixel as a sequence feature for classification [8]. Capsule networks (CapsNets) encode the properties of visual entities into vector outputs, which can extract high-level discriminative features for HSI spectral–spatial classification [9]. Containing the embeddings of topological structure, graph convolutional networks (GCNs) characterize the underlying structures of RS data for image classification [10]. Recently, to explore long-range relationships between pixels, the vision transformers (ViTs) specific to RS data are proposed to further boost the classification performance [11], [12]. In addition to the above model-based methods, some machine learning strategies have also been investigated for RS image classification, such as transfer learning [13], attention mechanism [14], and sparse representation [15]. These supervised DL models are data-driven ones, whose classification accuracies rely heavily on sufficient training samples. However, there are a limited number of annotated samples; solving the scarcity and imbalance of annotated samples is one major problem for the RS image land cover classification.

Many semisupervised methods have been proposed to utilize both labeled and unlabeled samples for classification. The graph-based learning is a typical semisupervised learning pattern for RS images. The joint graph structure is learned from multimodal RS images; thereafter, graph-based label propagation is conducted for semisupervised classification [16], [17]. The nonlocal graph structures are constructed from labeled and unlabeled samples; then, graph convolutional layers are employed to extract deep graph features for classification [18]. As a novel semisupervised classification scheme, the generative adversarial network

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(GAN) has also been employed for RS image classification, in which faked samples generated by the generator and labeled samples are utilized for classification [19]. Due to the high-dimensional and complex characteristics of RS data, there are great difficulties in generating realistic samples. Another important research direction in solving the small sample classification problem is the metalearning paradigm [20]. The feature extraction networks are trained on precollected labeled samples to learn such a metric space in which samples from the same category are clustered, while those from different classes are separated; then, trained networks are transferred to other datasets for feature extraction [21]. The classification accuracy of this few-shot learning paradigm is limited, and the training process also needs numerous precollected annotated samples.

Although there are many classification approaches in the RS field, current classification paradigms cannot effectively solve the most outstanding problem, that is, there are numerous unlabeled multimodal samples but a limited number of annotated samples. Self-supervised learning (SSL), as a special type of representation learning, learns meaningful feature representations from unlabeled data by designing a pretext task. By comparing the views derived from samples, contrastive learning aims to learn such a latent space in which similar sample pairs are clustered, while dissimilar ones are separated away, thus learning both invariant and discriminative features. Recently, contrastive learning has made great breakthroughs in the computer vision community, and its fine-tuning accuracy has caught up with or even surpassed the supervised classification paradigm.

However, there are multimodal RS images in the same observed scenes, and current contrastive learning models are proposed for natural images, which cannot utilize complementary information contained in multimodal data for feature learning. Some multimodal fusion models have been investigated for land cover classification, the most direct pattern is to stack multimodal features and then feed them into a classifier for classification [22]. The encoder–decoder architectures are utilized to integrate high-level knowledge information contained in multimodal RS data [23], [24]. The multibranch network structure is also used to extract multimodal features in subnetworks for joint classification [25]. The attention mechanism and the postprocessing approach are employed for multimodal RS images collaborative classification as well [26], [27]. These multimodal fusion models are suitable for the supervised classification paradigm, which cannot be directly utilized in the SSL scheme.

In this article, we investigated the multitask contrastive learning architecture for multimodal RS images, which not only constructs multiple views from multimodal RS data but also learns robust feature representations with multitask contrastive learning. The proposed multimodal SSL method makes full use of rich information for feature learning, which is more suitable for multimodal RS image classification, and experimental results confirm the superiority of the proposed SSL architecture. More specifically, the main contributions are summarized as follows.

1) We bring forward a multiview learning strategy to construct several complementary views from multimodal RS data, and in each view, we employ contrastive learning to learn high-level feature representations in a self-supervised manner.
2) To learn more robust feature representations from multimodal data, we propose the multitask learning architecture for contrastive learning, in which we utilize a parameter sharing strategy to train the base feature representation network.
3) We train the feature representation network on different source datasets and utilize the trained model for unsupervised feature extraction, and learned features are more beneficial to the land cover classification task.

The remainder of this article is organized as follows. Related work regarding unsupervised feature extraction of RS images is introduced in Section II. In Section III, we elaborate on the self-supervised feature representation (SSFR) architecture and few-shot classification method for multimodal RS images. We analyze the experimental results and main parameters in Section IV. Finally, in Sections V and VI, this article is discussed and summarized.

II. Related Work

Unsupervised learning models have the potential to learn meaningful feature representations from unlabeled data, and researchers have conducted several valuable explorations of unsupervised feature extraction.

A. Autoencoder-Based Approaches

The stacked autoencoder (SAE) extracts latent features by minimizing the reconstruction loss, which has been employed as an unsupervised feature extraction method for RS data [28], [29], [30]. Some modified autoencoder-based models have also been utilized for HSI unsupervised feature extraction, such as the deep residual con-deconv network [31] and 3-D convolutional autoencoder [32]. To further improve the feature learning performance, the compact and discriminative SAE (CDSAE) imposes a local Fisher discriminant regularization on each SAE [33]. The latent relationship guided stacked sparse autoencoder (LRSSAE) combines the stacked sparse autoencoder with the graph regularization and the superpixel constraints for unsupervised feature extraction [34].

Although autoencoder-based methods yield impressive classification performance, they often involve complex training processes, which also needs numerous labeled samples in the classification stage. Thus, traditional autoencoder-based methods have limitations in extracting meaningful key features to solve the small sample classification problem.

B. Self-Supervised Algorithms

In recent years, contrastive learning methods have shown great advantages for SSL in the computer vision field. However, there are several differences between computer vision and RS. The first one is that each pixel in the RS image has a geological meaning. In computer vision, every

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image has three bands, and it is difficult to interpret the image from an individual pixel. However, in the RS community, especially in the field of hyperspectral RS, each pixel has many spectral values, which brings unique advantages for image interpretation. The second difference is that scenes in RS images are more complex than those in computer vision. For example, RS images contain many different ground objects, and these different categories need to be considered simultaneously. The third one is that there are multimodal RS images within the same observed scenes, and how to effectively utilize these multimodal images is another major problem. Therefore, to utilize contrastive learning for learning key features from RS data, we must consider these differences to investigate contrastive learning models specific to multimodal RS images.

Recently, there are some self-supervised algorithms specific to RS data processing and analysis. The typical contrastive learning method for HSI is to construct two augmented views from two groups of spectral bands, which are employed for contrastive learning, and these learned features are employed for land cover classification [35], [36], [37]. However, these methods only utilize spatial information for feature learning, and SSL methods using both spatial and spectral features are also investigated for HSI classification [38], [39]. The swin transformer is used as a base feature extractor to exploit multiscale semantic representations of HSI as well [40]. The contrastive learning paradigm has also been investigated for RS images scene classification [41], PolSAR land cover classification [42], change detection [43], and so on. In addition to contrastive learning, low-rank learning [44], adversarial adaptation [45], and knowledge distribution [46] are also employed for RS image SSL.

It is a fact that there are multimodal RS images within the same observed scenes, and these images have rich and complementary information for downstream tasks [47]. To utilize complementary information contained in multimodal RS data, two observed views are constructed from hyperspectral and LiDAR data for contrastive learning [48]. However, current SSL algorithms generally construct two observed views from RS data for contrastive learning, which have limitations in fully utilizing multimodal RS data for self-supervised feature learning. To address the above issues, we investigate the multitask contrastive learning architecture to learn more discriminative feature representations from multimodal RS images for the following land cover classification task.

III. Methodology

In this section, we elaborate on the proposed multimodal RS image SSFR and few-shot land cover classification approach. This method mainly contains two stages, i.e., self-supervised feature learning and few-shot classification. In the feature learning procedure, multiple complementary views are constructed from multimodal RS data, each view is augmented separately for contrastive learning, and multitask learning strategy is also employed to learn more robust feature representations from multiple views. After the self-supervised feature learning stage, the samples of the same class aggregate into clusters in the latent space, and the samples of different categories are separated away; thereafter, the support vector machine (SVM) is utilized for few-shot classification.

A. Multitask Contrastive Learning

Contrastive learning is a popular and effective methodology for self-supervised feature learning [49]. Contrastive learning extracts meaningful features by comparing augmented samples that come from the same view, and the training process does not require any labeled information. The contrastive loss function aims to make similar pairs stay close while separating dissimilar ones in the feature space.

In multimodal RS images, there are multiple views from different spectral bands or different modalities, and these views contain a wealth of complementary information; thus, we propose the multimodal contrastive learning architecture, as shown in Fig. 1. The training process of the proposed self-supervised feature learning model includes four steps: constructing multiple views of the same observed scenes from multimodal images, augmenting each view to get a pair of samples, inputting each pair of augmented samples into the feature representation networks $f(\cdot)$ and $g(\cdot)$ to generate abstract feature $\boldsymbol{h}$ and output feature $\boldsymbol{z}$, and calculating the contrastive loss using output feature $\boldsymbol{z}$.

For hyperspectral data, each spectral band can be regarded as a single view because different spectral bands reflect specific reflection characteristics of ground objects. However, there is great information redundancy between neighboring spectral bands, and numerous views also complicate the training process. To maintain the complementarity between different views, we divide all spectral bands of HSI into several groups, and in each group, we adopt a principal component analysis (PCA) operation to extract principal features. The first three principal components are treated as an independent view to control the amount of calculation to an appropriate level while preserving information as much as possible. As for the VHR image, we directly employ it as an independent view. We normalize all views obtained from multimodal RS data to maintain the balance between different views.

Let $\boldsymbol{x}_n \in \mathbb{R}^{H\times W\times 3}$ represent one sample of the nth view constructed from multimodal RS images, in which H and W denote height and width, respectively. Two separate data augmentation operators sampled from the same family of augmentation are applied to each sample to obtain a pair of correlated samples $\boldsymbol{x}_{ni}$ and $\boldsymbol{x}_{nj}$. The inputs $\boldsymbol{x}_{ni}$ and $\boldsymbol{x}_{nj}$ are mapped to the latent vectors $\boldsymbol{h}_{ni}$ and $\boldsymbol{h}_{nj}$ through a nonlinear function $f(\cdot)$. The latent embedding is defined such that the high-dimensional feature vector represents the high-level semantic information of each sample. Then, a multilayer perceptron $g(\cdot)$ is employed to transform latent abstract features $\boldsymbol{h}_{ni}$ and $\boldsymbol{h}_{nj}$ into $\boldsymbol{z}_{ni}$ and $\boldsymbol{z}_{nj}$ for feature condensation and pruning. Thereafter, $\boldsymbol{z}_{ni}$ and $\boldsymbol{z}_{nj}$ are employed to calculate the contrastive loss.

In the training process for the nth view, a minibatch of N samples is randomly selected from total samples; thus, there are $2N$ augmented samples. The contrastive loss is calculated on pairs of $2N$ augmented samples; if two augmented samples

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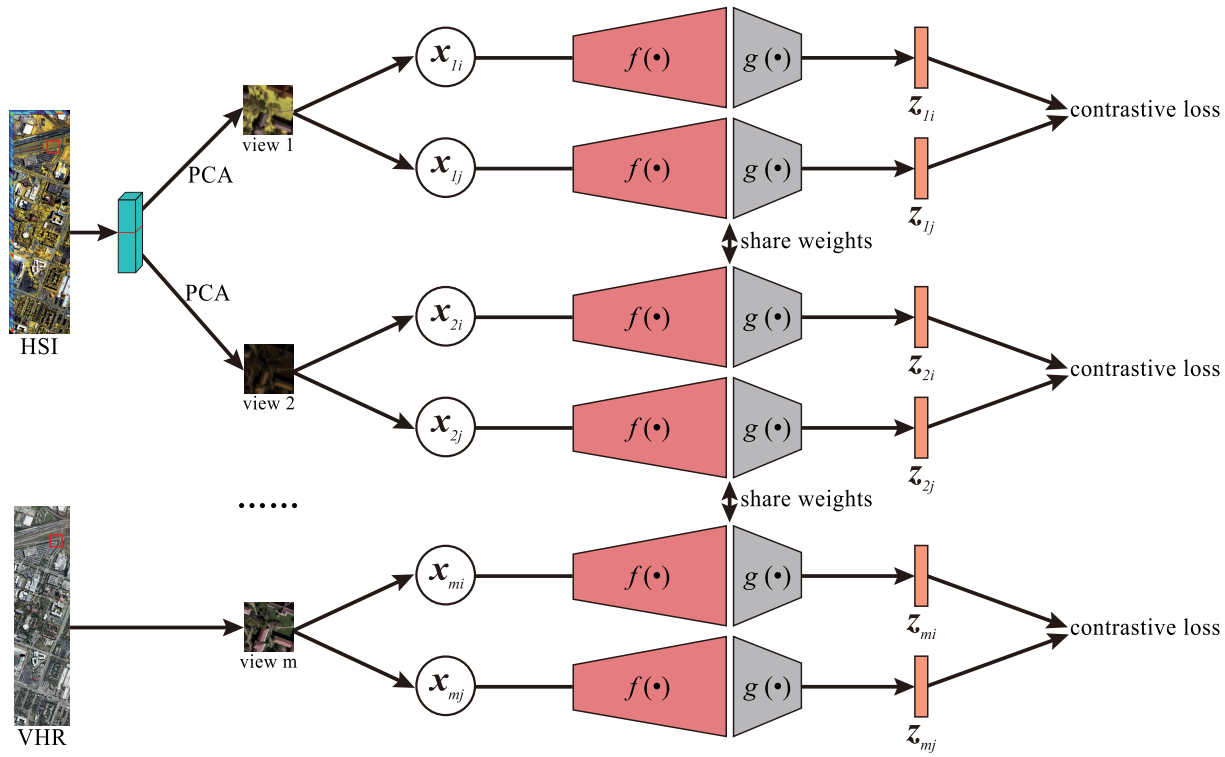

Fig. 1. Schematic architecture of proposed SSFR model for multimodal RS image. The input of this framework is multimodal RS data, multiple complementary views are constructed by applying the PCA on different groups of spectral bands from HSI, and the VHR image serves as a single view. Each view is augmented separately, and then, each pair of augmented samples are employed for contrastive learning. The feature representation networks are weights shared between all views to learn robust feature representations.

are augmented from the same sample, they are defined as a positive pair; otherwise, it is defined as a negative pair. The loss function for a pair of positive augmented samples is defined as follows:

$$\ell_{i,j}^{n} = -\log \frac{\exp\left(\text{sim}\left(z_{ni}, z_{nj}\right)/\tau\right)}{\sum_{k=1}^{2N} l_{[k\neq i]} \exp\left(\text{sim}\left(z_{ni}, z_{nj}\right)/\tau\right)} \tag{1}$$

in which $l_{[k\neq i]} \in \{0, 1\}$ is an indicator function evaluating to 1 iff $k \neq i$ and τ denotes a temperature parameter. (z_{ni}, z_{nj}) is the cosine similarity between output vectors z_{ni} and z_{nj}. There are m views constructed from multimodal RS images, and each view is employed for contrastive learning. To utilize multiple views for learning more robust representations, we make $f(\cdot)$ and $g(\cdot)$ share the same weights between different views. Therefore, the total contrastive loss on a minibatch for m views is defined as follows:

$$\ell = \sum_{n=1}^{m} \ell^{n} \tag{2}$$

where m denotes the number of views constructed from multimodal RS images and ℓ^n represents the contrastive loss of the nth view. Because contrastive loss is defined as judging pairs of augmented samples, which makes the feature representation network learn both invariant and discriminative features, the multitask learning strategy further makes the base feature representation network learn more robust semantic representations. Through this self-supervised feature learning, the learned features of the same category gather into clusters in the latent space, while features of different classes are separated away from each other.

B. Network Architecture

The data augmentation operation stochastically generates two related views of the same sample, which plays a critical role in contrastive learning. The feature representation network learns meaningful information by judging the augmented samples whether derived from the same view. In the proposed SSL architecture, two data augmentations (i.e., random Gaussian blur and random color distortions) specific to RS data are utilized for data augmentation.

The function $f(\cdot)$ extracts high-level feature vectors from images, which could employ various typical deep network architectures, such as ResNet, DenseNet, VGG, and Xception. Due to its easy training and strong feature extraction capabilities, residual networks have become common models to extract high-level features in the computer vision field [50]. Original residual networks are proposed for image classification, in which the final fully connected layer coverts high-dimensional vectors into categories. Thus, we remove the final fully connected layer and employ the variant of residual networks as the nonlinear feature extraction function $f(\cdot)$. Considering the calculation amount and feature extraction capability, we adopt a variant of ResNet-50 as the base feature extraction network. The main structures of the employed feature extraction network are shown in Table I.

After the feature embeddings of the base feature extractor, a multilayer perceptron with two fully connected layers $g(\cdot)$

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TABLE I
MAIN STRUCTURES OF THE BASE FEATURE EXTRACTION NETWORK

	Stage 0	Stage 1	Stage 2	Stage 3	Stage 4
Parameters	3×3, 64	$\begin{bmatrix} 1 \times 1, 64 \\ 3 \times 3, 64 \\ 1 \times 1, 256 \end{bmatrix} \times 3$	$\begin{bmatrix} 1 \times 1, 128 \\ 3 \times 3, 128 \\ 1 \times 1, 512 \end{bmatrix} \times 4$	$\begin{bmatrix} 1 \times 1, 256 \\ 3 \times 3, 256 \\ 1 \times 1, 1024 \end{bmatrix} \times 6$	$\begin{bmatrix} 1 \times 1, 512 \\ 3 \times 3, 512 \\ 1 \times 1, 2048 \end{bmatrix} \times 3$

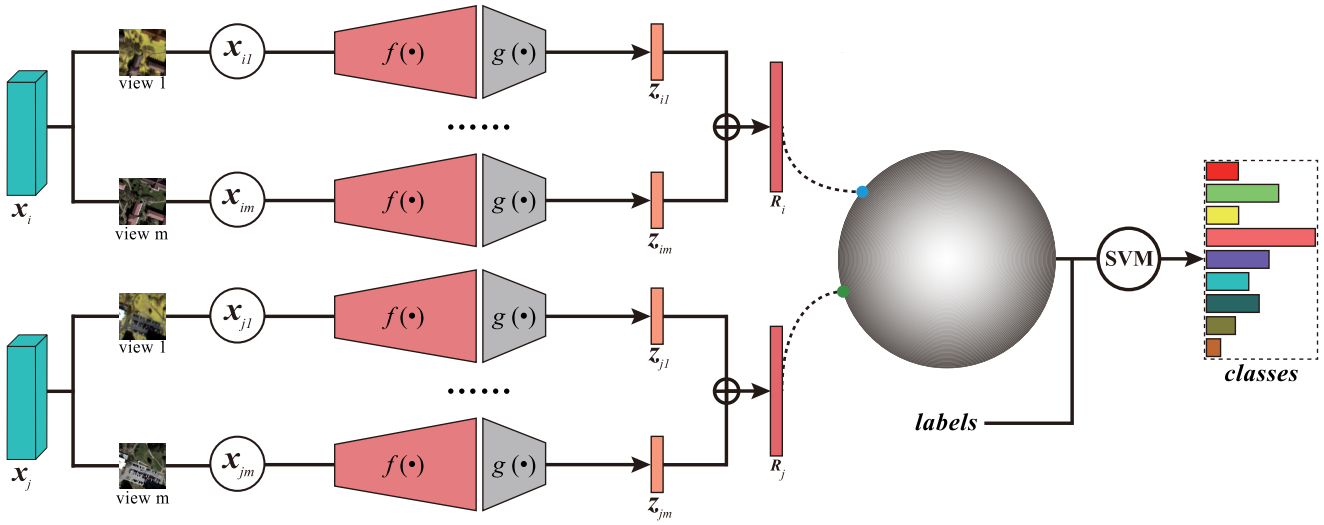

Fig. 2. Schematic architecture of few-shot land cover classification for multimodal RS image. All samples in multimodal RS data pass through the trained feature extractor, and multiple feature vectors corresponding to multiple views are concatenated to get the final feature representation. Samples of the same class are clustered in the learned latent space, while samples of different categories are separated. The SVM is applied to classify samples using a few annotated samples.

is employed to prune features. This operation reduces the original feature dimension (i.e., 2048) to an appropriate value (e.g., 64, 128, and 256), which could reduce the amount of calculation while selecting representative features for classification. In the proposed contrastive learning architecture, we set the dimension of output features as 128.

C. Few-Shot Classification

After the self-supervised feature learning, the learned feature representations are more conducive to the following classification task, and the few-shot classification architecture is shown in Fig. 2. In the classification phase, the trained feature extraction networks serve as the unsupervised feature extractor. All samples of the multimodal RS dataset pass through this feature extractor to output corresponding feature vectors. Note that all views are directly input into the feature extractor, and they no longer perform the data augmentation operation. Since different views contain complementary information, the extracted feature vectors of all views are concatenated as the final feature representation. The spectral features are concatenated with learned features to further improve the discriminative capability of features. In the learned feature space, the feature representations of samples are more discriminative. Thus far, the SVM classifies the learned feature representations with a few labeled samples.

IV. EXPERIMENTS

In this section, we report feature extraction and land cover classification results in this research, and the main parameters are also analyzed. We conduct our experiments on five groups of RS datasets including two HSIs: two multimodal RS datasets and different source datasets. The SSFR model is implemented within the PyTorch library, and other comparative approaches employ corresponding original experimental environments. All experiments are carried out on a workstation equipped with the Intel Core i9-7900X and three Nvidia GeForce RTX 2080 Ti graphics cards.

A. Implementation Details

In comparative experiments, we evaluate the feature learning capability and classification performance both quantitatively and qualitatively. The main evaluation coefficients [i.e., overall accuracy (OA), average accuracy (AA), and kappa coefficient (κ)] and individual class accuracy are employed to assess the classification results quantitatively. The feature visualization and classification maps are also utilized to evaluate the feature representation and classification performance from the visual perspective. In the training procedure of self-supervised feature learning, the number of epochs is set to 100, and the batch size is set to 64. In the supervised classification process, regularization parameter C and kernel parameter γ are selected by the cross-validation strategy in the range of $\{2^{-2}, 2^{-1}, \ldots, 2^{7}\}$ [51].

We employ several state-of-the-art classification approaches for comparison to evaluate the classification performance of the SSRF model, and these comparative methods include supervised and semisupervised classification methods using

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TABLE II
LAND COVER CATEGORIES AND CORRESPONDING SAMPLES OF THE SALINAS HYPERSPECTRAL DATASET

No.	Class name	Train	Test	Percent(%)
C1	Brocoli_green_weeds_1	5	2009	0.25
C2	Brocoli_green_weeds_2	5	3726	0.13
C3	Fallow	5	1976	0.25
C4	Fallow_rough_plow	5	1394	0.35
C5	Fallow_smooth	5	2678	0.18
C6	Stubble	5	3959	0.13
C7	Celery	5	3579	0.14
C8	Grapes_untrained	5	11271	0.04
C9	Soil_vinyard_develop	5	6203	0.08
C10	Corn_senesced_green_weeds	5	3278	0.15
C11	Lettuce_romaine_4wk	5	1068	0.47
C12	Lettuce_romaine_5wk	5	1927	0.26
C13	Lettuce_romaine_6wk	5	916	0.55
C14	Lettuce_romaine_7wk	5	1070	0.47
C15	Vinyard_untrained	5	7268	0.07
C16	Vinyard_vertical_trellis	5	1807	0.28
	Total	80	50550	0.16

traditional and DL models, and recent unsupervised feature learning approaches.

SVM [52] is a traditional supervised classification approach that has been widely employed for the RS image classification task, and the input of SVM is the original spectral vector. The regularization parameter and kernel parameter are optimized by the cross-validation strategy.

Transductive SVM (TSVM) [53] is a semisupervised method, which utilizes both labeled and unlabeled spectral vectors for training, and the radial basis function (RBF) kernel is also employed.

The contextual deep CNN (CDCNN) [6] is a supervised DL approach, which constructs the deep neural network using 2-D CNN and residual learning structure.

The spectral–spatial residual network (SSRN) [7] combines the residual learning and 3-D CNN to build an end-to-end supervised model, in which spectral and spatial features are extracted by corresponding convolution kernels.

The 3-D GAN (3DGAN) [19] is a semisupervised classification method based on GAN. The generator tries to generate fake samples, which are combined with labeled samples as input for the discriminator network.

The deep few-shot learning (DFSL) [21] is a transfer learning-based approach, which trains a deep feature extraction network on numerous precollected annotated samples. The trained feature extractor is then transferred to other datasets for feature extraction; thereafter, the SVM classifier utilizes extracted features for land cover classification.

The deep multiview learning (DMVL) [35] is a self-supervised feature learning model, in which two views are constructed from HSI and these views are augmented separately for contrastive learning. The ResNet-50 is utilized as the base feature extractor, and other parameters are set the same as the corresponding paper.

B. Experiment 1: Hyperspectral Data

Salinas: The Salinas dataset was gathered by the AVIRIS sensor over Salinas Valley, California. This hyperspectral

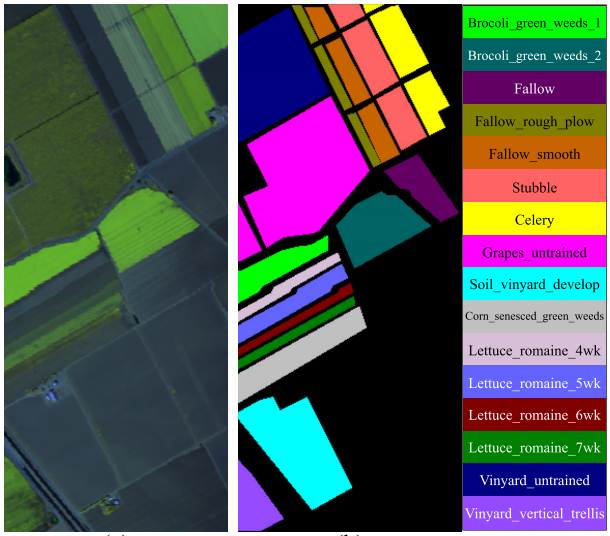

Fig. 3. Visualization of the Salinas hyperspectral dataset. (a) Pseudocolor image of the hyperspectral data. (b) Thematic distribution map.

TABLE III
LAND COVER CATEGORIES AND CORRESPONDING SAMPLES OF THE HANCHUAN HYPERSPECTRAL DATASET

No.	Class name	Train	Test	Percent(%)
C1	Strawberry	5	44735	0.01
C2	Cowpea	5	22753	0.02
C3	Soybean	5	10287	0.05
C4	Sorghum	5	5353	0.09
C5	Water spinach	5	1200	0.42
C6	Watermelon	5	4533	0.11
C7	Greens	5	5903	0.08
C8	Trees	5	17978	0.03
C9	Grass	5	9469	0.05
C10	Red roof	5	10516	0.05
C11	Gray roof	5	16911	0.03
C12	Plastic	5	3679	0.14
C13	Bare soil	5	9116	0.05
C14	Road	5	18560	0.03
C15	Bright object	5	1136	0.44
C16	Water	5	75401	0.01
	Total	80	257530	0.03

dataset has a spatial size of 512 × 217 pixels, and the corresponding spatial resolution is 3.7 m. There are 224 spectral bands ranging from 400 to 2500 nm. This dataset contains 16 land cover categories, and corresponding samples are listed in Table II. Fig. 3 shows the pseudocolor image and corresponding ground-truth map.

HanChuan: The HanChuan dataset covers a farming area in Hanchuan, Hubei, China, which was collected by the Headwall Nano-Hyperspectral imaging sensor [54]. There are 274 spectral bands ranging from 400 to 1000 nm. The spatial size of this imagery is 1217 × 303 pixels, and the spatial resolution is about 0.109 m. The cover area of this data is a typical agricultural scene, which contains 12 different land cover categories. The detailed ground-truth categories and corresponding samples are reported in Table III. The visualization of this dataset is given in Fig. 4.

1) Quantitative Evaluation: We employ main evaluation coefficients to evaluate the classification performance

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TABLE IV
CLASSIFICATION RESULTS WITH DIFFERENT APPROACHES ON THE SALINAS HYPERSPECTRAL DATASET

Class No.	SVM [52]	TSVM [53]	CDCNN [6]	SSRN [7]	3DGAN [19]	DFSL [21]	DMVL [35]	SSFR
C1	97.87	99.89	44.81	97.19	99.20	96.38	99.75	99.19
C2	98.75	99.72	81.37	97.67	99.31	96.86	99.65	99.66
C3	81.52	87.57	90.47	91.74	92.82	96.86	99.39	97.32
C4	97.31	97.76	90.14	95.77	95.45	69.81	70.65	85.15
C5	95.55	97.17	86.79	95.86	98.13	97.09	83.52	99.73
C6	99.63	99.89	98.21	99.92	99.66	93.07	90.56	92.37
C7	94.50	95.34	95.34	96.18	99.57	93.98	99.98	100
C8	68.83	77.91	76.36	80.51	74.30	91.92	98.46	99.75
C9	98.19	98.76	97.27	98.67	98.67	97.89	99.75	99.81
C10	79.67	86.38	82.26	91.93	92.65	97.38	98.86	99.34
C11	75.51	68.88	61.56	90.33	94.86	96.21	99.81	96.59
C12	86.06	93.48	88.92	94.85	97.38	95.00	99.50	95.95
C13	84.52	96.53	86.70	96.11	98.69	75.78	81.90	86.30
C14	77.51	98.13	91.43	95.74	90.82	86.84	94.31	91.52
C15	51.63	62.04	51.62	56.47	59.33	67.45	89.81	96.59
C16	80.02	87.78	96.82	96.54	86.65	96.78	99.94	99.78
OA	80.29 ±2.15	86.12 ±1.91	76.05 ±2.63	86.86 ±1.63	87.02 ±1.60	88.63 ±0.47	94.63 ±0.52	**97.30 ±0.43**
AA	85.44 ±1.54	90.45 ±1.24	82.50 ±2.03	92.22 ±1.63	92.34 ±1.21	90.58 ±0.54	94.11 ±0.64	**96.19 ±0.42**
κ	78.17 ±2.32	84.61 ±2.08	73.55 ±2.79	85.42 ±1.82	85.53 ±1.91	87.79 ±0.59	94.02 ±0.58	**96.99 ±0.37**

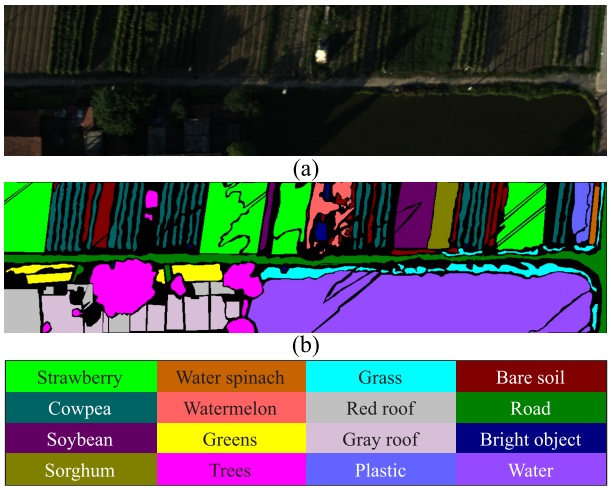

Fig. 4. Visualization of the HanChuan hyperspectral dataset. (a) Pseudocolor image of the hyperspectral data. (b) Thematic distribution map.

quantitatively, and Tables IV and V report classification results obtained by different approaches on Salinas and HanChuan datasets, respectively. Note that, for all approaches, only five annotated samples per class are utilized for classification. To increase the reliability of experimental results, we conduct ten trials for each classification experiment, and we randomly select the training samples in each trail. From the experimental results, we can come to the following conclusions.

First, semisupervised classification methods have higher classification accuracies than supervised ones. Because semisupervised classification models utilize both labeled and unlabeled samples for training, which can learn more discriminative features under limited labeled samples. In two groups of classification experiments, semisupervised methods (e.g., TSVM and 3DGAN) generally have higher classification accuracies than supervised ones (e.g., SVM and SSRN).

Second, few-shot learning models have better classification performance than supervised DL approaches. Because we employ very few annotated samples (i.e., five samples per class) for training, the supervised DL models (e.g., CDCNN and SSRN) encounter the overfitting problem, which seriously affects the classification performance. The SVM method even has higher classification accuracy than the CDCNN model.

Third, the self-supervised feature learning approaches have better classification performance than methods using original features. Behind this phenomenon is that feature learning methods can learn meaningful features from original data. For example, DFSL and DMVL approaches generally have higher classification accuracies than TSVM and 3DGAN.

Furthermore, the proposed self-supervised feature learning method is an effective unsupervised feature extraction pattern. Using the same number of labeled samples, the proposed method achieves the best classification accuracy in terms of OA, AA, kappa coefficient (κ), and individual class accuracy. The reason is that we utilize multiple views constructed from hyperspectral data for contrastive learning, thereby learning more meaningful features for the following classification task.

Last but not least, comparing the classification accuracies of two hyperspectral datasets, we can find that the accuracies of the HanChuan dataset obtained by different approaches are lower. The reason is that the structure of this dataset is more complex. However, by employing multiple views constructed from hyperspectral data for contrastive learning, the classification accuracy of this dataset is significantly improved. The average OA of SSFR reaches 85.90%, which is 4.62% higher than the DMVL model (i.e., 81.28%). Therefore, using multiple views for contrastive learning is important for self-supervised feature learning, especially for datasets with more complex structures.

2) Classification Maps: In addition to evaluating the classification results using main evaluation coefficients, we also employ classification maps to assess the classification

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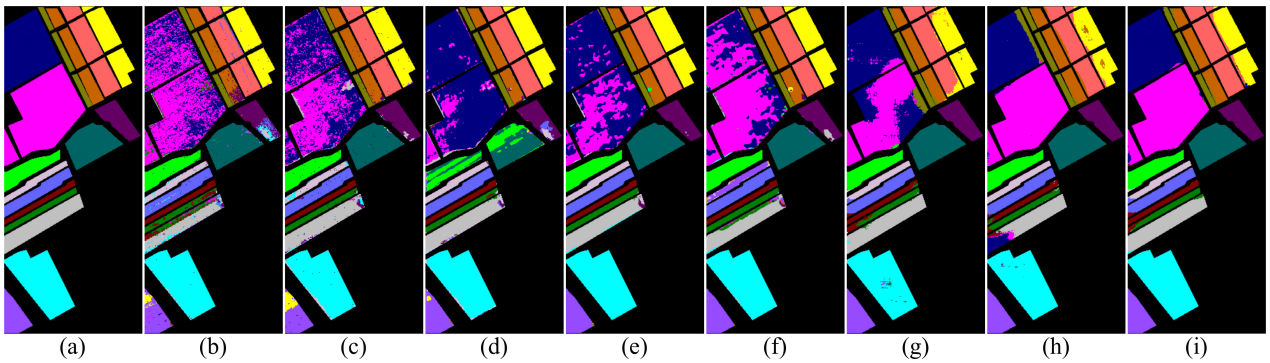

Fig. 5. Classification maps obtained by different approaches on the Salinas hyperspectral dataset. (a) Thematic distribution map. (b) SVM (OA: 80.29%). (c) TSVM (OA: 86.12%). (d) CDCNN (OA: 76.05%). (e) SSRN (OA: 86.86%). (f) 3DGAN (OA: 87.02%). (g) DFSL (OA: 88.63%). (h) DMVL (OA: 94.63%). (i) SSFR (OA: 97.30%).

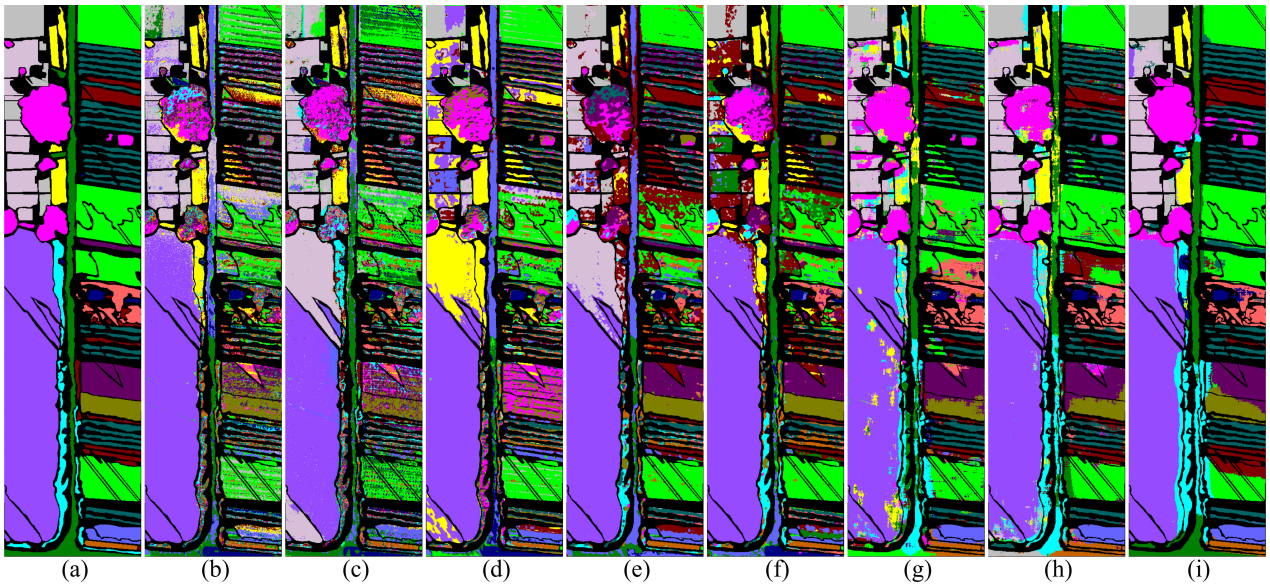

Fig. 6. Classification maps obtained by different approaches on the HanChuan hyperspectral dataset. (a) Thematic distribution map. (b) SVM (OA: 50.77%). (c) TSVM (OA: 56.38%). (d) CDCNN (OA: 52.66%). (e) SSRN (OA: 62.97%). (f) 3DGAN (OA: 70.67%). (g) DFSL (OA: 74.82%). (h) DMVL (OA: 81.28%). (i) SSFR (OA: 85.90%).

performance from a visual perspective. The classification maps obtained by different classification approaches on two benchmark datasets are shown in Figs. 5 and 6, respectively. In these figures, each color corresponds to a specific land cover category, and ground-truth maps are also displayed for better comparison. When comparing classification maps in (b)–(i) with ground-truth maps, we find that classification maps achieved by semisupervised classification models and unsupervised feature extraction methods have less noisy pixels. The classification maps obtained by the SSFR method are more homogeneous, which have the fewest noisy pixels. This is because the proposed SSL architecture employs multiple views to extract key features. Thus, the lightweight classification model utilizes extracted features for better classification performance with very few labeled samples.

3) Feature Visual Analysis: To further assess the feature learning capability of different approaches, we employ the distributed stochastic neighbor embedding (t-SNE) algorithm to display the distribution of extracted features in the 2-D space [55]. The visualization results of two benchmark datasets are shown in Fig. 7, in which different colors stand for different land cover categories. In (a) and (e), we can observe that the original spectral features of different categories are confused. The transfer learning-based DFSL and the DMVL based on contrastive learning increase the distance between different classes of samples to a relatively large value, and it can be considered that the separability of different categories is increased. As for (d) and (h), the distances between different classes are the largest, and samples of the same category are clustered together, which means that features extracted by SSFR are more discriminative compared with other feature learning methods.

C. Experiment 2: Multimodal Remote Sensing Data

Trento: This dataset consists of hyperspectral and LiDAR data, whose coverage is a typical rural scene. This dataset contains 166 × 600 pixels with a spatial resolution of 1 m. There are 63 spectral bands within the 402.89–989.09-nm spectrum range, and the corresponding spectral resolution is 9.2 nm. There are six thematic categories within the coverage of this dataset, and land cover classes and the number of samples are reported in Table VI; furthermore, Fig. 8 gives

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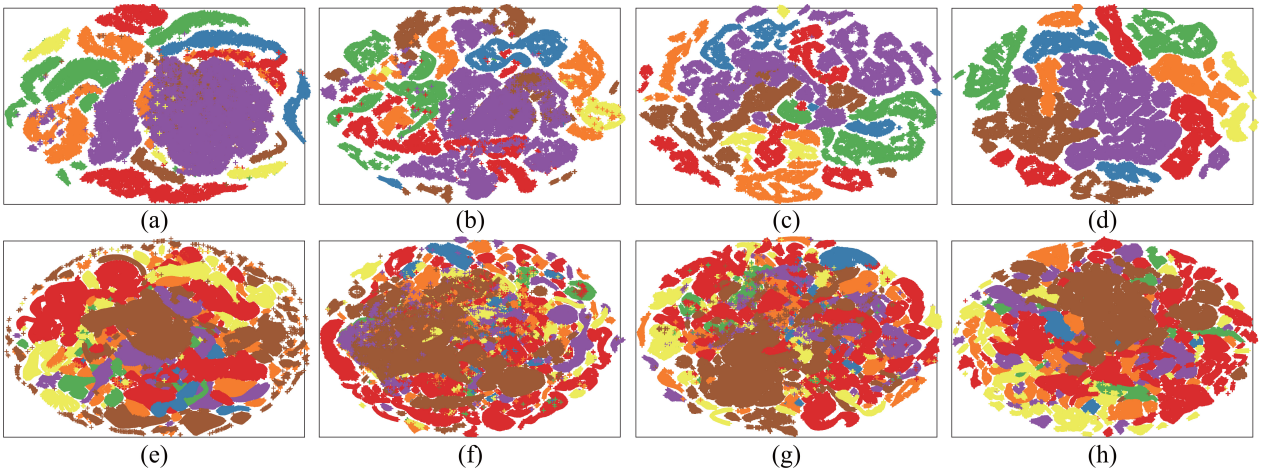

Fig. 7. t-SNE visualization of features obtained by different models on two benchmark datasets. The Salinas hyperspectral dataset: (a) original spectral features, (b) features extracted by DFSL, (c) features extracted by DMVL, and (d) features extracted by SSFR. The HanChuan hyperspectral dataset: (e) original spectral features, (f) features extracted by DFSL, (g) features extracted by DMVL, and (h) features extracted by SSFR.

TABLE V
CLASSIFICATION RESULTS WITH DIFFERENT APPROACHES ON THE HANCHUAN HYPERSPECTRAL DATASET

Class No.	SVM [52]	TSVM [53]	CDCNN [6]	SSRN [7]	3DGAN [19]	DFSL [21]	DMVL [35]	SSFR
C1	84.70	82.90	90.86	91.12	93.22	81.19	90.65	97.47
C2	46.27	58.38	67.94	71.79	89.18	91.89	89.67	89.64
C3	31.24	35.71	22.81	48.27	55.52	72.38	83.44	91.08
C4	33.55	42.51	15.19	66.62	77.90	95.19	88.05	81.23
C5	8.95	14.66	16.31	10.72	15.84	31.61	31.62	65.66
C6	8.76	9.91	10.30	30.40	42.90	34.21	62.02	77.28
C7	33.40	62.76	15.31	46.78	41.41	27.59	51.23	73.36
C8	57.63	60.77	47.02	78.75	95.64	70.71	82.70	86.64
C9	16.95	30.34	12.73	61.06	65.10	24.27	37.45	38.78
C10	40.45	48.74	61.81	82.17	99.67	87.17	66.53	73.48
C11	36.20	32.63	25.12	48.02	98.82	64.24	83.19	83.76
C12	11.07	24.23	11.00	12.86	27.15	78.08	90.98	99.12
C13	23.20	34.83	13.36	29.54	19.24	65.49	48.13	58.99
C14	62.27	71.50	73.33	74.54	53.41	66.09	80.12	88.05
C15	10.97	17.78	20.88	43.75	30.06	48.85	73.60	54.40
C16	99.30	93.41	86.70	98.89	98.13	99.27	99.44	98.41
OA	50.77 ±3.82	56.38 ±1.41	52.66 ±2.08	62.97 ±1.29	70.67 ±1.56	74.82 ±0.86	81.28 ±1.35	**85.90 ±0.91**
AA	37.81 ±1.73	45.07 ±0.93	36.92 ±1.82	55.96 ±1.45	62.70 ±1.71	64.89 ±0.91	72.43 ±1.26	**78.58 ±0.87**
κ	44.73 ±3.86	50.65 ±1.38	45.96 ±2.17	59.11 ±1.36	66.14 ±1.58	70.95 ±1.06	78.35 ±1.31	**83.67 ±1.04**

the visualization of HSI, LiDAR-derived rasterized DSM, and ground truth of this dataset.

Houston 2018: The Houston 2018 dataset was gathered over the university of Houston campus, which is provided by the 2018 IEEE GRSS Data Fusion Competition [56]. We employ the training portion of the whole hyperspectral data, and the corresponding VHR image is resampled to be consistent in the spatial resolution. The spatial size of this dataset is 601 × 2384 pixels with a 1-m ground sampling distance (GSD). There are 48 spectral bands ranging from 380 to 1050 nm in hyperspectral imagery. This dataset covers 20 land cover categories, and Table VII shows the detailed information of samples. The illustrations of the pseudocolor image of hyperspectral data, the VHR image, and the ground-truth map are given in Fig. 9.

1) Quantitative Evaluation: Tables VIII and IX report detailed classification results in terms of main evaluation coefficients and individual class accuracy obtained by

TABLE VI
LAND COVER CATEGORIES AND CORRESPONDING SAMPLES OF THE TRENTO DATASET

No.	Class name	Train	Test	Percent(%)
C1	Apple Trees	5	4034	0.12
C2	Buildings	5	2903	0.17
C3	Ground	5	479	1.04
C4	Wood	5	9123	0.05
C5	Vineyard	5	10501	0.05
C6	Roads	5	3174	0.16
	Total	30	30214	0.10

different classification approaches on Trento and Houston 2018 datasets. For the Trento dataset, we employ the invariant attribute profiles (IAPs) [57] to extract three profiles from LiDAR-derived DSM, which are treated as a single view. For a fair comparison, we concatenate the VHR or extracted profiles to hyperspectral data as input for other comparative

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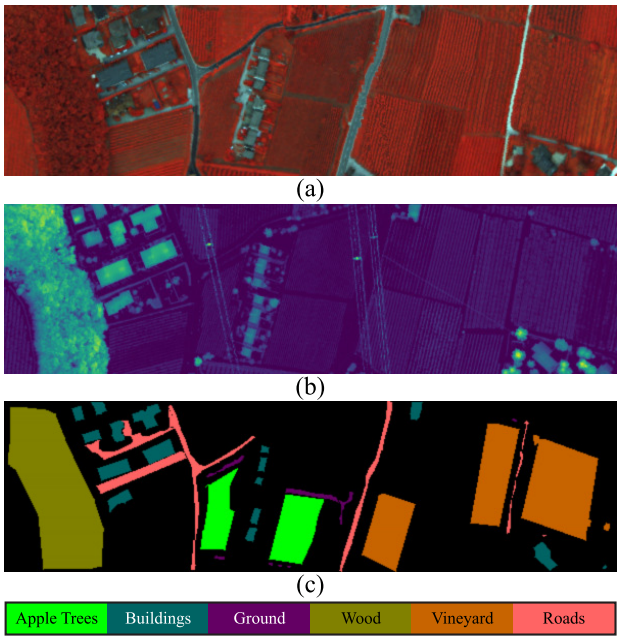

Fig. 8. Visualization of the Trento dataset. (a) Pseudocolor image of the hyperspectral data. (b) LiDAR-derived rasterized DSM. (c) Thematic distribution map.

TABLE VII
LAND COVER CATEGORIES AND CORRESPONDING SAMPLES OF THE HOUSTON 2018 DATASET

No.	Class name	Train	Test	Percent(%)
C1	Healthy grass	5	9799	0.05
C2	Stressed grass	5	32502	0.02
C3	Artificial turf	5	684	0.73
C4	Evergreen trees	5	13588	0.04
C5	Deciduous trees	5	5048	0.10
C6	Bare earth	5	4516	0.11
C7	Water	5	266	1.88
C8	Residential buildings	5	39762	0.01
C9	Non-residential buildings	5	223684	0.002
C10	Roads	5	45810	0.01
C11	Sidewalks	5	34002	0.01
C12	Crosswalks	5	1516	0.33
C13	Major thoroughfares	5	46358	0.01
C14	Highways	5	9849	0.05
C15	Railways	5	6937	0.07
C16	Paved parking lots	5	11475	0.04
C17	Unpaved parking lots	5	149	3.36
C18	Cars	5	6578	0.08
C19	Trains	5	5365	0.09
C20	Stadium seats	5	6824	0.07
	Total	100	504712	0.02

approaches, and we also conduct ten trials for each experiment. From these tables, we can get the following conclusions.

First, we can see that unsupervised feature extraction is important for classification, especially for a more complex dataset. Because the Houston 2018 dataset covers 20 urban categories and there are 51 bands, classification methods using original data cannot extract discriminative features under limited training samples; thus, classification accuracies obtained by different methods are generally low. Unsupervised feature extraction approaches can learn meaningful information from unlabeled data, and these learned features facilitate the final classification task.

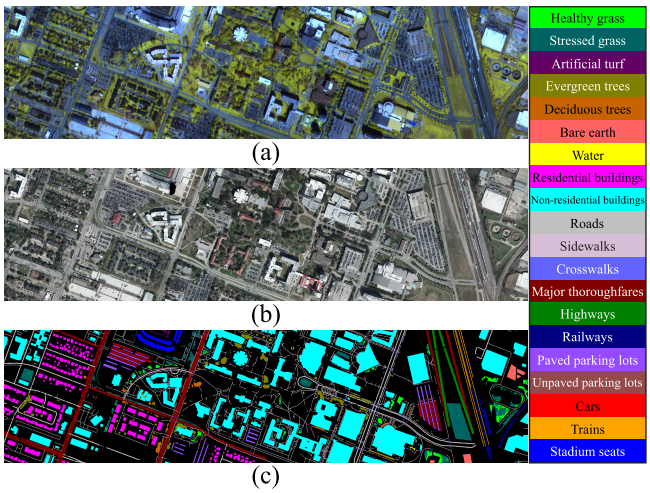

Fig. 9. Visualization of the Houston 2018 dataset. (a) Pseudocolor image of the hyperspectral data. (b) VHR image. (c) Thematic distribution map.

Second, multimodal RS data (e.g., hyperspectral and VHR data) have rich complementary information for land cover classification. The proposed method constructs multiple views from multimodal RS images, and we employ multitask learning strategy to extract robust features from these views. The classification accuracies obtained by the proposed method are higher than other comparative methods, which proves that SSFR could extract more discriminative features from multimodal RS data for classification.

2) Classification Maps: We also employ classification maps to appraise the classification performance from a visual perspective, and classification maps obtained by different approaches on two benchmark datasets are shown in Figs. 10 and 11. When comparing classification maps with ground-truth maps, we can find that classification maps achieved by other unsupervised feature extraction methods have misclassified areas. Classification maps obtained by the SSFR method are the most homogeneous. This is because the proposed self-supervised feature learning architecture extracts more discriminative features from multimodal data and employs extracted features for better classification.

D. Experiment 3: Different Source Data

HyRANK: The HyRANK dataset is developed in the framework of the ISPRS scientific initiatives [58]. There are five HSIs in this dataset, in which the training set includes two HSIs (i.e., Dioni and Loukia) and the validation set contains three HSIs (i.e., Erato, Nefeli, and Kiriki). In our experiments, we employ the Dioni data and three validation HSIs for self-supervised feature learning, and the Dioni data are employed to evaluate the classification performance. The spatial size of the Dioni data is 250 $\times$ 1376 pixels, and there are 176 spectral bands. There are 14 ground objects in the HyRANK dataset, while the Dioni data covers 12 categories. The detailed information of samples in the Dioni data is reported in Table X. Fig. 12 gives the visualization of the HyRANK dataset.

1) Quantitative Evaluation: The main evaluation coefficients and individual class accuracy obtained by different

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TABLE VIII

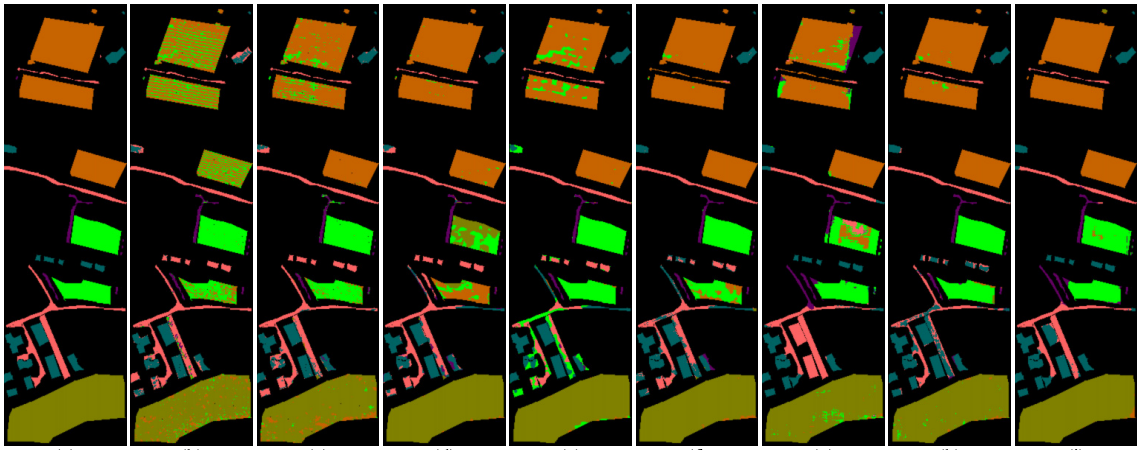
CLASSIFICATION RESULTS WITH DIFFERENT APPROACHES ON THE TRENTO DATASET

Class No.	SVM [52]	TSVM [53]	CDCNN [6]	SSRN [7]	3DGAN [19]	DFSL [21]	DMVL [35]	SSFR
C1	49.97	82.40	94.42	90.06	87.56	78.89	91.83	98.82
C2	83.77	73.58	72.99	90.11	86.68	83.35	83.44	89.42
C3	70.86	74.74	75.37	63.88	98.22	26.86	78.94	57.66
C4	99.22	99.16	89.88	99.65	98.64	99.81	99.84	99.90
C5	87.88	96.03	86.33	96.89	96.21	95.75	97.24	99.85
C6	80.98	73.46	79.71	80.67	53.60	69.22	86.46	95.85
OA	80.13 ±0.11	88.81 ±0.20	87.12 ±0.33	92.85 ±0.20	90.43 ±0.21	87.04 ±0.09	94.31 ±0.15	**96.50 ±0.15**
AA	78.78 ±0.22	83.23 ±0.38	83.12 ±0.25	86.88 ±0.13	86.82 ±0.28	75.65 ±0.10	89.63 ±0.17	**90.25 ±0.12**
κ	74.19 ±0.14	85.20 ±0.26	82.44 ±0.37	90.48 ±0.27	87.25 ±0.27	82.97 ±0.12	92.43 ±0.16	**95.35 ±0.13**

TABLE IX

CLASSIFICATION RESULTS WITH DIFFERENT APPROACHES ON THE HOUSTON 2018 DATASET

Class No.	SVM [52]	TSVM [53]	CDCNN [6]	SSRN [7]	3DGAN [19]	DFSL [21]	DMVL [35]	SSFR
C1	52.77	59.28	30.10	36.15	50.07	17.96	21.53	31.74
C2	73.15	78.99	46.30	58.09	83.12	41.44	75.01	77.52
C3	17.50	4.96	3.71	24.03	48.10	4.57	12.93	26.45
C4	73.87	75.19	51.73	56.41	69.17	42.40	52.80	49.65
C5	10.52	15.47	10.02	12.93	23.72	22.60	14.18	20.34
C6	8.28	19.83	8.24	10.30	20.15	19.77	31.36	67.72
C7	73.71	13.14	16.26	76.69	38.16	1.65	8.36	33.43
C8	45.36	51.87	38.18	39.99	40.39	91.36	95.28	93.95
C9	87.56	93.01	86.55	93.33	91.47	94.41	93.92	92.50
C10	24.50	33.04	19.82	32.39	30.83	50.34	41.84	55.75
C11	21.05	33.56	25.51	26.71	38.85	17.96	18.99	28.94
C12	1.41	1.90	1.47	1.33	7.10	4.37	3.41	25.04
C13	31.44	42.04	33.69	23.92	42.43	86.32	90.06	73.58
C14	26.61	29.93	17.18	9.10	34.82	62.41	60.89	67.52
C15	19.87	34.07	18.39	13.24	42.15	74.06	97.59	87.70
C16	20.70	23.03	14.28	11.36	25.84	58.83	59.59	76.50
C17	1.14	2.33	0.45	0.44	2.48	0.66	1.23	21.52
C18	12.62	21.53	9.43	18.44	14.89	32.15	35.01	50.48
C19	13.00	28.84	12.43	14.48	14.73	82.24	78.56	96.72
C20	35.50	23.52	20.12	31.99	41.35	31.63	98.41	81.64
OA	32.02 ±3.08	40.02 ±3.66	24.78 ±2.75	30.22 ±2.68	40.06 ±1.54	56.09 ±2.42	61.10 ±1.59	**68.11 ±0.85**
AA	32.53 ±2.28	34.28 ±1.75	23.19 ±2.80	29.57 ±2.54	37.99 ±1.40	41.86 ±2.36	49.55 ±1.46	**57.93 ±0.92**
κ	26.45 ±2.36	34.26 ±2.97	20.01 ±2.69	24.70 ±2.53	34.19 ±1.52	47.49 ±2.51	52.06 ±1.28	**60.71 ±1.00**



(a) (b) (c) (d) (e) (f) (g) (h) (i)

Fig. 10. Classification maps obtained by different approaches on the Trento dataset. (a) Thematic distribution map. (b) SVM (OA: 80.13%). (c) TSVM (OA: 88.81%). (d) CDCNN (OA: 87.12%). (e) SSRN (OA: 92.85%). (f) 3DGAN (OA: 90.43%). (g) DFSL (OA: 87.04%). (h) DMVL (OA: 94.31%). (i) SSFR (OA: 96.50%).

classification methods on the Dioni data are reported in Table XI. Because the proposed feature learning architecture is a self-supervised method, in which labeled samples are not required during the training process. We employ three validation HSIs (i.e., Erato, Nefeli, and Kiriki) and Dioni data as training data in the feature learning procedure.

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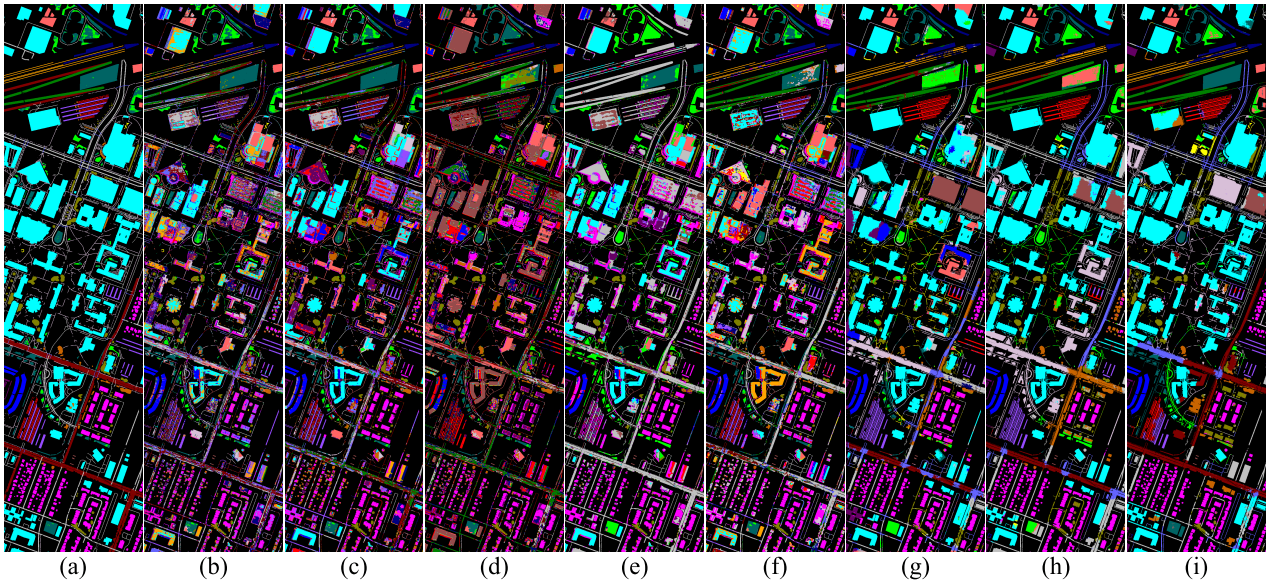

Fig. 11. Classification maps obtained by different approaches on the Houston 2018 dataset. (a) Thematic distribution map. (b) SVM (OA: 32.02%). (c) TSVM (OA: 40.02%). (d) CDCNN (OA: 24.78%). (e) SSRN (OA: 30.22%). (f) 3DGAN (OA: 40.06%). (g) DFSL (OA: 56.09%). (h) DMVL (OA: 61.10%). (i) SSFR (OA: 68.11%).

TABLE X

LAND COVER CATEGORIES AND CORRESPONDING SAMPLES OF THE DIONI DATASET

No.	Class name	Train	Test	Percent(%)
C1	Dense Urban Fabric	5	1262	0.40
C2	Mineral Extraction Sites	5	204	2.45
C3	Non-Irrigated Arable Land	5	614	0.81
C4	Fruit Trees	5	150	3.33
C5	Olive Groves	5	1768	0.28
C6	Coniferous Forest	5	361	1.39
C7	Dense Sclerophyllous Vegetation	5	5035	0.10
C8	Sparse Sclerophyllous Vegetation	5	6374	0.08
C9	Sparsely Vegetated Areas	5	1754	0.29
C10	Rocks and Sand	5	492	1.01
C11	Water	5	1612	0.31
C12	Coastal Water	5	398	1.26
	Total	60	20024	0.30

The trained feature representation network is employed as a feature extractor for the Dioni data; thereafter, the SVM is used to classify the extracted features. From classification results in Table XI, we can observe that, by utilizing other source data, the proposed SSFR model learns more discriminative feature representations, and these features promote the following land cover classification task. The classification accuracies obtained by SSFR are higher than DFSL, which is also trained on other source datasets, which indicates that the proposed method learns more meaningful features from other source datasets.

2) Classification Maps: The resulting classification maps are also employed to evaluate the classification performance qualitatively. Since labeled samples account for a small proportion of the Dioni data, we employ the full classification map for better visual evaluation. The classification maps obtained by different approaches on the Dioni data are shown in Fig. 13. When comparing the classification maps, we can observe that the classification maps achieved by feature learning-based methods have fewer noisy pixels. The classification map obtained by the SSFR method is the most homogeneous. The classification map obtained by DMVL gives a false category to the Water class, but the SSFR could correctly classify the Water category, which means that the investigated feature learning approach extracts more discriminative information from other source datasets.

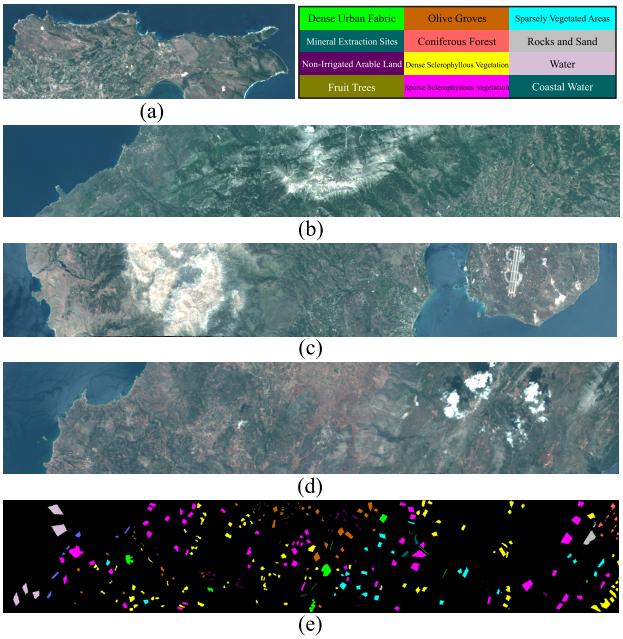

Fig. 12. Visualization of the HyRANK dataset. (a) Pseudocolor image of the Nefeli data. (b) Pseudocolor image of the Erato data. (c) Pseudocolor image of the Kirki data. (d) Pseudocolor image of the Dioni data. (e) Thematic distribution map of the Dioni data.

E. Parameter Sensitivity Analysis

Some hyperparameters in the proposed SSFR model affect the representation capability of extracted features, which, in turn, influences the land cover classification performance.

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TABLE XI
CLASSIFICATION RESULTS WITH DIFFERENT APPROACHES ON THE DIONI DATASET

Class No.	SVM [52]	TSVM [53]	CDCNN [6]	SSRN [7]	3DGAN [19]	DFSL [21]	DMVL [35]	SSFR
C1	28.37	46.73	28.16	31.92	36.74	60.00	61.35	91.91
C2	11.99	61.39	60.19	93.02	51.28	19.34	63.35	63.35
C3	36.87	75.29	22.49	48.43	76.47	62.05	53.85	45.94
C4	21.11	39.51	19.80	21.85	30.64	30.82	28.63	62.33
C5	20.97	40.45	38.15	95.66	65.79	61.50	68.12	90.31
C6	85.04	25.38	62.29	37.21	85.30	53.29	74.13	73.52
C7	94.52	80.16	86.52	83.73	96.09	92.77	75.90	80.11
C8	81.31	78.72	80.14	86.63	73.79	87.41	91.84	85.43
C9	62.80	67.20	48.90	49.48	64.39	61.48	79.24	80.11
C10	94.87	92.05	38.66	47.60	85.88	59.03	86.65	56.02
C11	97.10	94.35	89.79	98.39	99.32	97.80	99.86	99.90
C12	27.81	47.38	47.65	98.40	99.60	99.68	59.93	58.96
OA	57.43 ±2.32	63.26 ±1.40	58.92 ±2.82	71.18 ±1.77	75.28 ±1.36	73.88 ±2.03	76.81 ±1.65	**78.79 ±0.86**
AA	55.23 ±2.56	62.38 ±1.68	51.90 ±2.92	66.03 ±2.41	72.11 ±1.41	65.46 ±1.87	70.24 ±1.34	**73.99 ±1.02**
κ	49.94 ±2.63	56.41 ±1.45	51.76 ±2.92	64.95 ±1.92	68.97 ±1.68	70.92 ±1.91	72.17 ±1.58	**74.24 ±0.92**

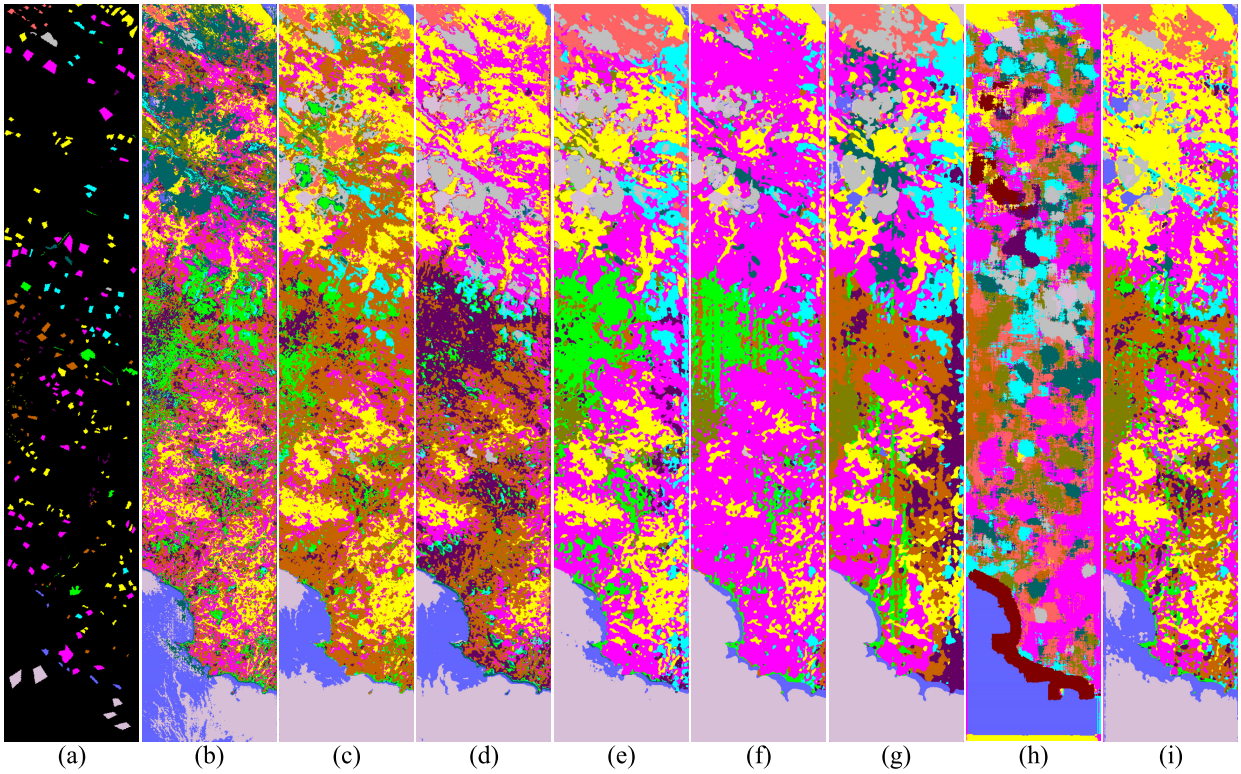

Fig. 13. Classification maps obtained by different approaches on the Dioni dataset. (a) Thematic distribution map. (b) SVM (OA: 57.43%). (c) TSVM (OA: 63.26%). (d) CDCNN (OA: 58.92%). (e) SSRN (OA: 71.18%). (f) 3DGAN (OA: 75.28%). (g) DFSL (OA: 73.88%). (h) DMVL (OA: 76.81%). (i) SSFR (OA: 78.79%).

This section analyzes the influence of main hyperparameters on feature extraction and classification performance.

1) Learning Rate: The learning rate controls the step of gradient descent in the self-supervised feature learning process, which is an important parameter affecting the feature learning and classification performance. In our experiments, we set the candidate set of learning rate as {0.0001, 0.0002, 0.0005, 0.001, 0.002, 0.005, 0.01, 0.02}, and we test the influence of this parameter on land cover classification accuracy. The overall accuracies using different learning rates on five benchmark datasets are reported in Fig. 14. From this figure, we can observe that classification accuracies with different learning rates vary greatly. Smaller learning rates correspond to more stable classification performance, and their classification accuracies are generally higher. According to the classification results on five benchmark datasets, the reasonable learning rates are set as 0.0001, 0.0002, 0.0001, 0.0001, and 0.0002, respectively.

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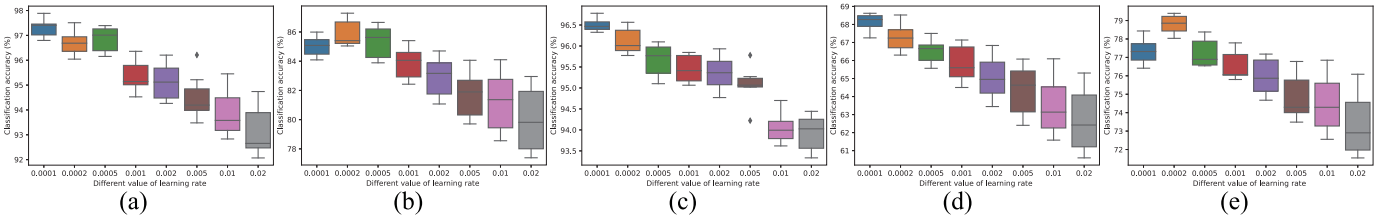

Fig. 14. Overall classification accuracies with different learning rates on five benchmark datasets. (a) Salinas dataset. (b) HanChuan dataset. (c) Trento dataset. (d) Houston 2018 dataset. (e) Dioni dataset.

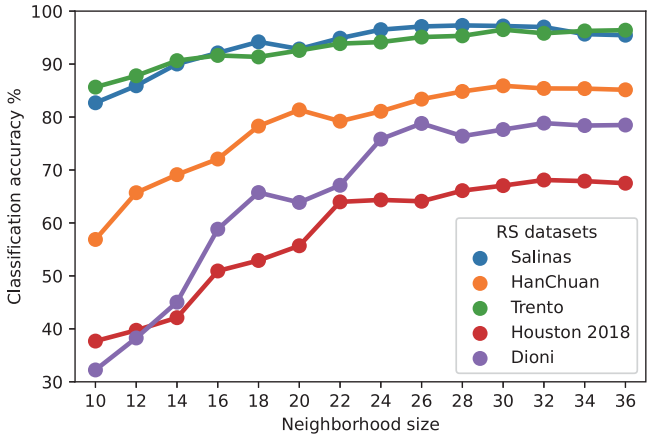

Fig. 15. Overall classification accuracies with different neighborhood sizes on five benchmark datasets.

2) Neighborhood Size: The neighborhood size controls the receptive field of samples, which also affects the feature learning performance. For the purpose of analyzing the influence of this parameter on classification accuracy, the candidate set of this parameter is set to be 10, 12, 14, 16, 18, 20, 22, 24, 26, 28, 30, 32, 34, and 36. The overall accuracies using different neighborhood sizes on five benchmark datasets are reported in Fig. 15. From this figure, we can come to the conclusion that too small and large neighborhood sizes influence the classification accuracies, and optimal neighborhood sizes for five datasets are 28, 30, 30, 32, and 26, respectively.

3) Feature Extraction Network: The base feature extraction network extracts high-level features from unlabeled samples, which is another parameter affecting the feature learning capability. To evaluate the influences of feature extraction network on classification performance, we employ several network backbones (i.e., VGG16 [59], ResNet-34, ResNet-50, ResNet-101, SE-ResNet-50 [60], and ViT-Base [61]) as the base feature extractor. The overall accuracies with different network backbones on five benchmark datasets are reported in Table XII. It could be observed that ResNet-50 generally has better classification performance on five RS datasets. Therefore, we employ the ResNet-50 as a base feature extractor to extract high-level features for land cover classification.

4) Number of Views: Because we employ multiple views in multimodal RS data for contrastive learning, the number of views influences the feature learning capability and calculation amount. The candidate set of extracted views is set to be $\{1, 2, 3, 4\}$, and we test the influences of this parameter

TABLE XII
CLASSIFICATION ACCURACIES ON FIVE BENCHMARK DATASETS USING DIFFERENT BACKBONE NETWORKS

	Salinas	HanChuan	Trento	Houston 2018	Dioni
VGG16	94.26	78.59	93.27	62.33	74.18
	0.42	0.55	0.77	0.71	0.58
ResNet-34	96.72	83.57	94.62	65.68	75.38
	0.31	0.52	0.63	0.62	0.43
ResNet-50	97.30	85.90	96.50	68.11	78.79
	0.43	0.91	0.15	0.85	0.86
ResNet-101	95.59	84.62	94.57	66.72	76.47
	0.67	0.89	0.21	0.46	0.76
SE-ResNet-50	96.29	86.26	95.91	68.22	77.73
	0.56	0.38	0.37	0.76	0.71
ViT-Base	95.88	85.35	95.51	66.54	76.11
	0.57	0.48	0.27	0.58	0.62

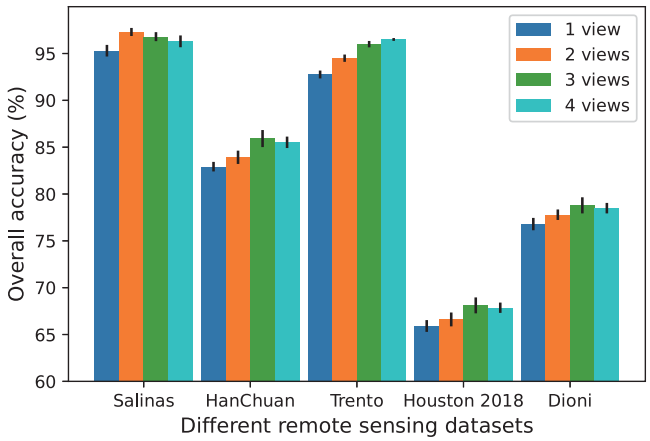

Fig. 16. Overall classification accuracies with different views on five benchmark datasets.

on classification performance. The overall accuracies using different views on five benchmark datasets are reported in Fig. 16. It can be observed that more views correspond to higher classification accuracy because more views have more complementary information, especially for the multimodal RS datasets (i.e., Trento and Houston 2018). According to the classification results, reasonable views for five benchmark datasets are 2, 3, 4, 3, and 3, respectively. These experimental results also confirm that more constructed views are conducive to contrastive learning.

5) Number of Labeled Samples: Since the investigated SSL model extracts key features for the following classification task, we also analyze the classification performance with different numbers of annotated samples. The overall accuracies with different numbers of labeled samples on five benchmark

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TABLE XIII
ABLATION STUDY WITH DIFFERENT CONTRASTIVE MODES ON FIVE BENCHMARK DATASETS

	Salinas			HanChuan			Trento			Houston 2018			Dioni		
	OA	AA	κ	OA	AA	κ	OA	AA	κ	OA	AA	κ	OA	AA	κ
Inner	97.30	96.19	96.99	85.90	78.58	83.67	96.50	90.25	95.35	68.11	57.93	60.71	78.79	73.99	74.24
	0.43	0.42	0.37	0.91	0.87	1.04	0.15	0.12	0.13	0.85	0.92	1.00	0.86	1.02	0.92
Cross	95.63	95.79	95.14	85.21	78.27	82.89	96.78	89.14	95.72	69.68	55.94	63.04	77.60	71.26	72.82
	0.35	0.54	0.46	0.77	0.95	0.82	0.10	0.27	0.14	0.76	0.58	0.86	0.68	1.26	0.79

TABLE XIV
ABLATION STUDY WITH DIFFERENT SAMPLING STRATEGIES ON FIVE BENCHMARK DATASETS

	Salinas			HanChuan			Trento			Houston 2018			Dioni		
	OA	AA	κ	OA	AA	κ	OA	AA	κ	OA	AA	κ	OA	AA	κ
Random	97.30	96.19	96.99	85.90	78.58	83.67	96.50	90.25	95.35	68.11	57.93	60.71	78.79	73.99	74.24
	0.43	0.42	0.37	0.91	0.87	1.04	0.15	0.12	0.13	0.85	0.92	1.00	0.86	1.02	0.92
Controlled	96.86	95.56	96.22	85.26	77.98	82.95	96.35	89.66	95.05	67.26	56.18	60.50	78.03	71.93	73.62
	0.33	0.54	0.37	0.61	0.78	0.70	0.26	0.45	0.31	0.72	0.81	0.63	0.74	0.81	0.90

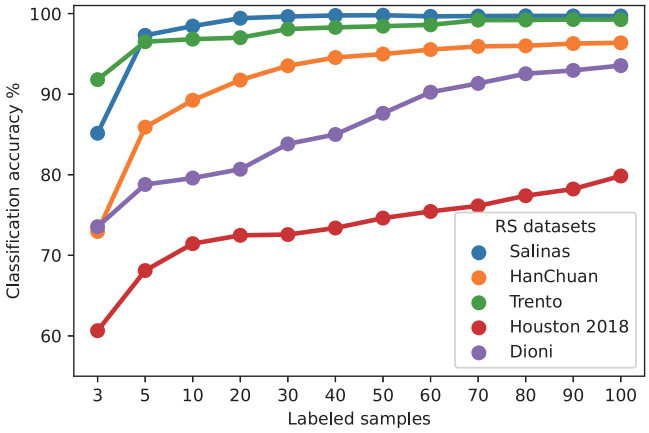

Fig. 17. Overall classification accuracies with different numbers of labeled samples on five benchmark datasets.

datasets are shown in Fig. 17. From this figure, we can observe that the model could achieve high classification accuracy even with a few labeled samples, i.e., three, five, and ten labeled samples per class, which indicates that this self-supervised feature learning architecture effectively extracts discriminative features for the classification task.

F. Ablation Studies

Since the proposed model benefits from several components, we conduct ablation studies to analyze how each component contributes to the classification performance.

1) Contrastive Learning Mode: Because there are multimodal RS data within the same observed scenes, the proposed method tries to employ complementary information for self-supervised feature learning. Multiple views are constructed from hyperspectral data or multimodal RS images, and contrastive learning is implemented for each view. In addition, the multitask learning strategy is employed to make full use of the complementary information between different views for learning more robust features. To evaluate the effectiveness of this architecture, we perform contrastive learning across different views, and the classification results using different contrastive learning modes are reported in Table XIII. From this table, we can observe that the contrastive learning operation performed on each view generally has higher classification accuracies, and the contrastive learning performed across different views achieves competitive classification results. The reason is that, although the proposed architecture performs contrastive learning within each view, the multitask learning strategy utilizes complementary information contained in multimodal data for feature learning.

2) Sampling Strategy: The spectral–spatial classification approaches suffer from the overlap between the training and test samples, which violates the principle of independence between training and test sets. To measure the influence of the sampling strategy on classification performance, we employ the controlled random sampling strategy for comparison [62]. The classification results using two sampling strategies are reported in Table XIV. It can be observed that the classification accuracies obtained by the two sampling strategies are comparable. The reason is that feature learning is an SSL paradigm that does not require any labeled information. In the classification phase, only five labeled samples per class are selected for classification, and the overlaps between training samples and testing samples for both methods are similar; thus, the classification results obtained by the two sampling strategies are comparable.

V. DISCUSSION

When using the DL models for multimodal RS images land cover classification, traditional models often have great difficulties in extracting discriminative features using very few labeled samples (e.g., five labeled samples per class). To address this problem, we design the self-supervised feature learning method, which learns meaningful features from numerous unlabeled samples. Using the extracted features, the few-shot classification model achieves better classification performance compared with several state-of-the-art models. The main reasons could be summarized as follows.

First, utilizing multiple views for contrastive learning is necessary. There are several views of the same observed scene in multimodal RS data, and these views contain rich information for the downstream task. The proposed

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self-supervised feature learning architecture utilizes each view for contrastive learning, and the multitask learning strategy also contributes to feature learning, which helps to learn robust feature representations.

Second, in the few-shot land cover classification, selecting labeled samples is important. In the few-shot classification procedure, we find that the classification accuracy varies greatly when selecting different annotated samples, which also leads to a large variance in classification accuracy. Because each sample contains different characteristics and there are a few labeled samples, thus, selected samples influence the classification performance greatly.

VI. Conclusion

In this study, we investigate a novel SSFR and few-shot land cover classification architecture for multimodal RS images, which has two main advantages. In the RS data land cover classification, there are multiple observed views that have rich and complementary information. Based on this fact, we construct multiple views from multimodal data, and each view is employed to learn discriminative information by contrastive learning. Then, to make full use of the complementary information contained in different views, we further propose the multitask learning strategy to learn more robust feature representations. Therefore, the proposed SSL architecture can extract more robust features for land cover classification. We carry out five groups of RS datasets for feature extraction and few-shot classification, and comparative experiments confirm the effectiveness and superiority of the proposed method.

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